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Optimum liquefaction fraction for boil-off gas reliquefaction system of semi-pressurized liquid CO_2 carriers based on economic evaluation

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ABSTRACT

This investigation studied the optimum reliquefaction fraction of a boil-off gas reliquefaction system for semi-pressurized liquid CO_2 carriers. To prevent the overpressure of cargo tanks, the boil-off gas should be properly treated in suitable ways: to be vented into the atmosphere or to be liquefied in a dedicated liquefaction system. Because the former accompanies the penalty of carbon tax and the latter requires energy consumption, there is an optimum reliquefaction fraction minimizing the total treatment expense. To estimate the optimum reliquefaction fraction, the total treatment expense was composed of the three terms: the reliquefaction expense, the emission penalty, and the benefit of CO_2 sale, which were expressed by the reliquefaction fraction and other variables. The reliquefaction expense increased rapidly with the reliquefaction fraction, while the other two showed a linear dependence. Even though each term of the total treatment expense varied with the crude price, the CO_2 sale price, and the carbon tax, the optimum reliquefaction fraction was found to stay within a narrow region, roughly between 95.7% and 98.2% for all the plausible scenarios for the variables.

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1. Introduction

The climate change arising from increases in CO_2 emissions has recently become a major concern throughout the world. The International Energy Agency (2006) stated that carbon capture and storage (CCS) should be considered as an important option to reduce green house gas emissions. Even though geological formations are indispensible for CCS, public anxiety over potential leaks due to stored CO_2 makes it difficult to locate onshore storage sites that are acceptable to the public. Naturally, seabed geological formations have arisen as promising storage candidates.

For transportation of the green house gas to the seabed sites, there are two options: ship-based and pipeline-based CCS. The latter is suitable to handle a large amount of CO_2 without intermediate storage. Its capital investment is much higher than the former when the distance from CO_2 source to storage is stretched (Svensson et al., 2004). Safety and economic considerations for pipeline transport have been well studied by many researchers (Skovholt, 1993; Gale and Davison, 2004). The ship transportation is more flexible than pipeline transportation in terms of transportation capacity and storage location. Ozaki and Ohsumi (2011) proposed the hub-type CCS by shuttle ships and socket buoys. They focused their attention on flexibility of ship-based CCS for plan change such as route, injection capacities, water depth, and transport distance. The FINNCAP project, though it was canceled recently, conducted a comparative feasibility study, leading to the conclusion of transportation by ships of 20,000 m^3 with approximately 2000 km of transportation distance (Rauramo, 2010). They assumed that ship transportation was suitable for long transportation distances and in the case of using CO_2 as enhanced oil recovery (EOR) with multiple offshore storage sites. Yoo et al. (2011) presented a new CCS system integration with CO_2 carrier and liquefaction process. They expected that CO_2 shipping for large amounts of CO_2 transport could be more competitive than pipeline transportation in the new total CCS chain consisting of capture, liquefaction, ship transportation, injection, and storage.

Cost effectiveness is the key to the realization of commercial scale CCS. In their extensive study on carrier-based transportation, Aspelund et al. (2006) considered the supply chain from onshore liquefaction to offshore unloading and analyzed the chain in view of cost, energy utilization, exergy and CO_2 emission. They indicated that the triple point was the optimum condition for the supply chain, and the energy requirement over the whole chain was 142 kWh/ton-CO_2 with a transportation cost of about 25 $/ton-$CO_2$ for a 1500 km roundtrip distance. Aspelund and Gundersen (2009) proposed a novel transport chain for liquefied CO_2, liquefied natural gas, and liquefied nitrogen including an offshore installation, a combined gas carrier, and an onshore integrated receiving terminal. In their concept, the cold exergy both in the offshore and onshore processes was utilized to make the whole chain energy-efficient

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Nomenclature

BOG	boil-off gas
B_S	benefit of CO_2 sale, \$/ton-BOG
CAPEX	capital expenditure
CCS	carbon capture and storage
COP	coefficient of performance
C_L	fuel cost, \$/kJ
C_P	penalty for unit CO_2 emission, \$/ton-CO_2
C_S	CO_2 sale price, \$/ton-CO_2
EOR	enhanced oil recovery
E_T	total treatment expense, \$/ton-BOG
E_P	penalty for CO_2 emission, \$/ton-BOG
JT	Joule–Thomson
MDO	marine diesel oil
$\dot{m}_B$	mass flow rate of generated BOG, kg/h
$\dot{m}_L$	mass flow rate of liquefied BOG, kg/h
$\dot{m}_V$	mass flow rate of vented BOG, kg/h
LCC	life-cycle cost
LCO_2	liquefied carbon dioxide
LNG	liquefied natural gas
LPG	liquefied petroleum gas
OPEX	operating expenditure
PUF	polyurethane form
tCO_2e	ton carbon dioxide equivalent
$\dot{W}_{tot}(x)$	total compressor work, kW
$w(x)$	specific work, kJ/kg

Greek symbol

k	proportional constant of fuel (see Eq. (8))

and cost-effective. For cost-effective CCS, EOR can be employed to store the captured CO_2. It can mitigate the life cycle cost of the CCS supply chain (Ferguson et al., 2009). Holt et al. (2009) estimated the CO_2 EOR potential in the UK and Norwegian oil field of the North Sea. They stated that the cost for CO_2 capture could be covered with the use of CO_2 as EOR for oil prices in the range of 90–100 \$/barrel.

To transport the greenhouse gas by ship, gaseous CO_2 should be liquefied to have a high density. Carbon dioxide exists as liquid on the saturated liquid curve bounded by the triple point (−56.6 °C, 5.2 bar) and the critical point (31.1 °C, 73.8 bar). The liquid density of CO_2 depends considerably on temperature and pressure. Along the saturated liquid curve, the density increases with decreasing temperature (equivalently decreasing pressure). Ozaki (1997) proposed a "Semi-Ref" type (−55 °C, 6 bar) of CO_2 carrier with capacity 22,000 m^3 consisting of four spherical tanks of 5500 m^3 unit capacity. He indicated that the maximum capacity for the onshore spherical storage tanks for liquid CO_2 near the triple point was 20,000 m^3 with about 34 m of diameter. Aspelund et al. (2006) developed an LPG/CO_2 semi-pressurized ship concept including liquefaction, intermediate storage, loading and offloading systems. They indicated that the triple point was the optimum transportation condition by analyzing cost, energy utilization, exergy and CO_2 emission.

One drawback with the low-temperature CO_2 carriers is the boil-off gas (BOG) generation. During a loaded voyage, heat ingress into the cargo tanks is caused by a significant temperature difference (about 60–90 °C), leading to a considerable amount of BOG. To prevent the overpressure of cargo tanks, the generated BOG should be properly treated.

There are two options to treat the generated BOG: atmospheric venting and reliquefaction. The former leads to cargo loss and greenhouse gas emission with negligible operation cost. Considering the current technology for the cargo containment system design for large liquefied gas carriers, the cargo loss is expected to reach several tons per hour for big CO_2 carriers. The penalty for the greenhouse gas emission will probably be implemented in the future. The latter option of BOG reliquefaction requires equipment, usually comprising turbo-machineries and heat exchangers. Additionally, it consumes electric energy generated at the expense of fuel oils, implying a considerable operating expenditure. Full reliquefaction seems to be uneconomical because the electric load would increase to liquefy the light-ends of nitrogen and oxygen. As a consequence, there should be an optimum reliquefaction fraction that minimizes the total treatment expense.

This study presents a BOG reliquefaction system for liquefying the BOG generated from semi-pressurized liquid CO_2 carriers and determines the optimum liquefaction fraction that minimizes the total treatment expense. Because most practical liquefaction systems use Joule–Thomson refrigeration systems to generate low temperature (Barrson, 1985), a cascade system with Joule–Thomson valves is considered in this study. The total treatment expense is estimated by defining the reliquefaction expense, the penalty for CO_2 emissions, and the benefit of recovered CO_2 through analyzing the CO_2 market and utilization of CO_2. The effect of the reliquefaction fraction on the total treatment expense is also delineated.

Table 1
Conditions of captured CO_2 and semi-pressurized liquid CO_2.

Item	Unit	Captured CO_2 (Sanpasertparnich et al., 2010)	Semi-pressurized liquid CO_2
Pressure	bar	1	6.5
Temperature	°C	30	−51
Composition	mole%		
CO_2		95.87	99.06
N_2		0.02	0.02
O_2		0.001	0.001
H_2O		4.11	0.917
SO_2		0.001	0.001

2. Design of BOG reliquefaction system

2.1. Target carrier and cargo conditions

The transportation of CO_2 at low temperature and pressure has some advantages though it is accompanied by a considerable operating cost for liquefaction or refrigeration. The saturated liquid density ranges from 1200 kg/m^3 at the triple point to 600 kg/m^3 at the critical point. Thus, given the cargo volume, the ship is able to transport almost two times as much at the triple point than at the critical point. In addition, the reduced design pressure has a positive impact on the material cost for the pressure vessels. For these reasons, the cargo condition is assumed to be 6.5 bar and −51 °C.

CO_2 is captured at CO_2 sources and delivered to the liquefaction plant, where it is liquefied into a semi-pressurized liquid at the cargo condition. The liquefied CO_2 is stored temporally and loaded into CO_2 carriers. The composition of the liquefied CO_2 is different that of the delivered gas. The composition of the liquefied CO_2 is obtained by compressing and refrigerating the captured gas from a coal-fired power plant (Sanpasertparnich et al., 2010) as shown in Table 1.

The currently available cargo capacity of CO_2 carriers, which serve to transport CO_2 as a commodity, ranges up to 1500 m^3 (Borresen, 2010). These carriers are small for the ship-based CCS purpose, and larger carriers are inevitable to rely on economies of scale. Carriers with larger cargo capacity seem to be achievable with current shipbuilding technology. Yoo et al. (2010) has designed CO_2 carriers, called the ECO_2 carriers, with a capacity up to 100,000 m^3.

Table 2
Characteristics of the cargo containment system for semi-pressurized liquid CO_2.

Item	Unit	Value
Cargo capacity	m^3	100,000
Cargo density	kg/m^3	1155
Vertical cylinder type tank	m × m	7 (D) × 25 (H)
No. of cargo tanks	EA	100
Initial filling level	%	95
Maximum filling limit	%	98

In this study, we assume that the cargo capacity is 100,000 m^3. Characteristics of the cargo containment system of the semi-pressurized liquid CO_2 carrier are shown in Table 2. The vertical cylindrical pressure vessels of IMO type C are installed. Several tanks are assumed to consist of a cargo group, which is an elementary unit for cargo handling operation; all the vessels in a cargo group are loaded and unloaded simultaneously. Thus, the level, pressure, and temperature in a cargo group are considered equal. The initial cargo filling level is 95%, allowing for the thermal expansion of the cargo but not exceeding the maximum filling limit of 98% dictated by IGC Code (International Maritime Organization, 2008). It is critical to keep the liquid level below the maximum limit for safety of cargo containment system.

3. BOG load

The BOG is generated due to the heat ingress into the low-temperature cargo containment system from the surrounding air and seawater. The BOG rate is directly proportional to the heat transfer rate from the surroundings to the cargo. The heat transfer rate largely depends on the heat transfer coefficient and heat transfer area as well as the temperature difference. Precise estimation of the BOG rate requires detailed information on the carrier such as the hull dimension, equipment arrangement, cargo tanks, insulation, environmental conditions, and voyage conditions. Because such information is not accessible for the moment, this study estimates a BOG rate based on the following reasoning.

The simplest form of the cargo containment system is shown in Fig. 1. The whole cargo containment system is insulated by the polyurethane form (PUF) on the inner hull plate. To be on the conservative side for the heat transfer rate, the temperature on the inner hull is assumed to be equal to the stored cargo temperature. Consequently, the heat transfer rate is controlled by the thickness of the insulation layer.

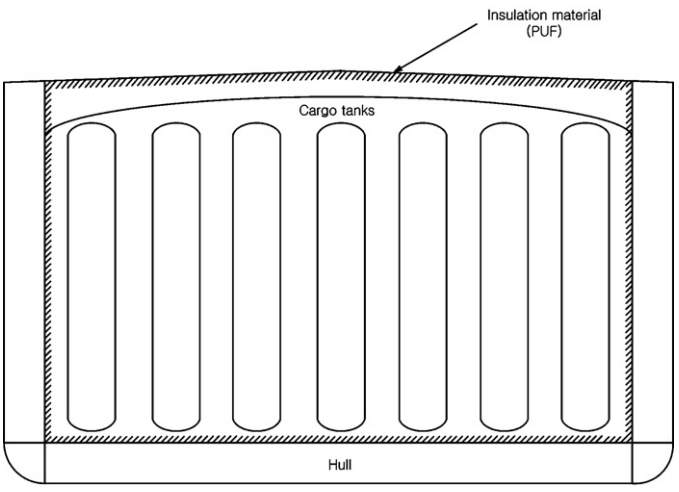


Fig. 1. Simple sketch of insulation system.

Table 3
Characteristics of BOG for reliquefaction.

Item	Unit	Value
BOG rate	%/day from total cargo	0.12
BOG load	kg/h	6000
BOG inlet pressure	bar	6.2
BOG inlet temperature	°C	−28.5
BOG composition	mole%	
CO_2		98.386
N_2		1.557
O_2		0.056

Typically, LNG carriers have a BOG rate in the range of 0.1–0.15% of the full cargo per day. Their cargo tanks are well thermally protected with special cryogenic insulation layers. On the contrary, semi-pressurized liquid CO_2 carriers do not have to be insulated as much as LNG carriers for the following reasons. First, the cargo temperature of the CO_2 carriers is far higher than that of the LNG carriers, roughly by 100 °C. It implies that, given the same insulation thickness for a similar cargo containment configuration, the heat ingress into CO_2 cargo tanks is roughly two thirds of that into LNG cargo tanks. Second, the CO_2 cargo is not as valuable as the LNG carriers. Usually, the LNG BOG is used for fuel for propulsion or reliquefied in a dedicated unit. Lastly, the CO_2 BOG may be vented, unlike the flammable LNG BOG. Venting of the former may be subject to greenhouse gas emission penalty, while that of the latter is strictly prohibited for the sake of safety as well as the emission penalty. As a consequence, it is not tempting to use as much insulation in the CO_2 carriers as in the LNG carriers. On the basis of these considerations, this study takes 0.12% of the full contents per day for the nominal BOG rate, which corresponds to an insulation thickness of 10 cm. With that BOG rate, the BOG load is estimated to be 5400 kg/h for CO_2 carriers with 100,000 m^3 of cargo tank capacity. Considering a margin for overload, the BOG load is assumed to be 6000 kg/h.

3.1. Design of BOG reliquefaction system

The pressure and temperature of the BOG into the reliquefaction unit are assumed to be 6.2 bar and −28.5 °C. The increase in the temperature is due to the heat ingress into the piping segments for the BOG transport from the cargo tanks to the reliquefaction unit. Even though they are insulated, heat ingress is inevitable. Characteristics of the BOG are summarized in Table 3.

The BOG composition is obtained on the basis of the phase equilibrium between the vapor (BOG) and liquid (semi-pressurized liquid CO_2). Mole fractions of the light components such as nitrogen and oxygen in the generated BOG are significantly increased from those in the stored liquid CO_2. Thus, the liquefaction temperature of the BOG should be lower than that of the semi-pressurized liquid CO_2 for full reliquefaction.

Commercial designs are available for BOG reliquefaction for conventional bulk gas carriers for LNG, LPG and crude (Cryostar, 2007; Hamworthy Ltd., 2007a,b,c). Two processes are available to generate a low-temperature cold source (Barrson, 1985): one uses expansion turbines, and the other uses expansion valves or Joule–Thomson valves. Though the former has higher efficiency due to the shaft work recovery from the expansion turbines, the latter is capable of handling two-phase flow with simpler equipment configuration, a lower footprint, more straightforward operation, and less maintenance demand. Consequently, the latter type is considered in this study.

The reliquefaction fraction highly depends on the condensation temperature, as shown in Fig. 2. Assuming that the lowest pressure of the Joule–Thomson refrigeration process is atmospheric in order for the air not to leak into the system, the condensation

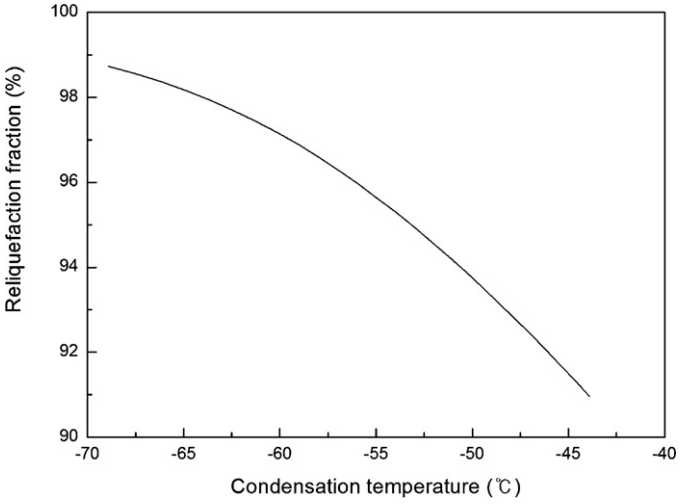


Fig. 2. Reliquefaction fraction with condensation temperature.

temperature governs the choice of the refrigerant. A Joule–Thomson refrigeration process using a pure refrigerant of propane has an upper bound in the reliquefaction fraction because the normal boiling temperature of propane is approximately −42 °C. Thus, the minimum cooling temperature of the BOG is approximately −37 °C with the temperature in the condenser approaching 5 °C. To accomplish the reliquefaction fraction over 98%, the BOG should be cooled down below −65.6 °C.

To achieve a sufficiently high reliquefaction fraction, ethylene is also employed as an additional refrigerant. This process, called a propane–ethylene JT refrigeration process, is of cascade cycle, which has two Joule–Thomson valves for the two single component refrigerants. The propane cycle plays the role of cooling both the ethylene and the BOG and removing heat to the ambient (sea water). The ethylene cycle fully condenses the BOG into liquid phase. The performance and feasibility of the process are analyzed on the following assumptions.

(1) The outlet temperature of the refrigerants from seawater heat exchangers is 40 °C.
(2) The minimum temperature approach in the heat exchangers is 5 °C.
(3) The polytropic efficiency of all compressors is 80%.
(4) The pressure drop in the heat exchangers is 10 kPa for condensing steam and 21 kPa otherwise.
(5) The refrigerants discharged from the evaporative heat exchangers are saturated vapor.
(6) The BOG discharged from the ethylene heat exchanger (before the expansion valve) is saturated liquid.

Aspen HYSYS (2009), the commercial program, was used to simulate the process with Peng-Robinson equation of state employed as a fluid package to calculate the properties of the BOG and the refrigerants. Fig. 3 shows the detailed features of the propane–ethylene JT refrigeration system containing three streams of BOG, propane, and ethylene. Propane passes through three compression stages and then exchanges heat with sea water to reject

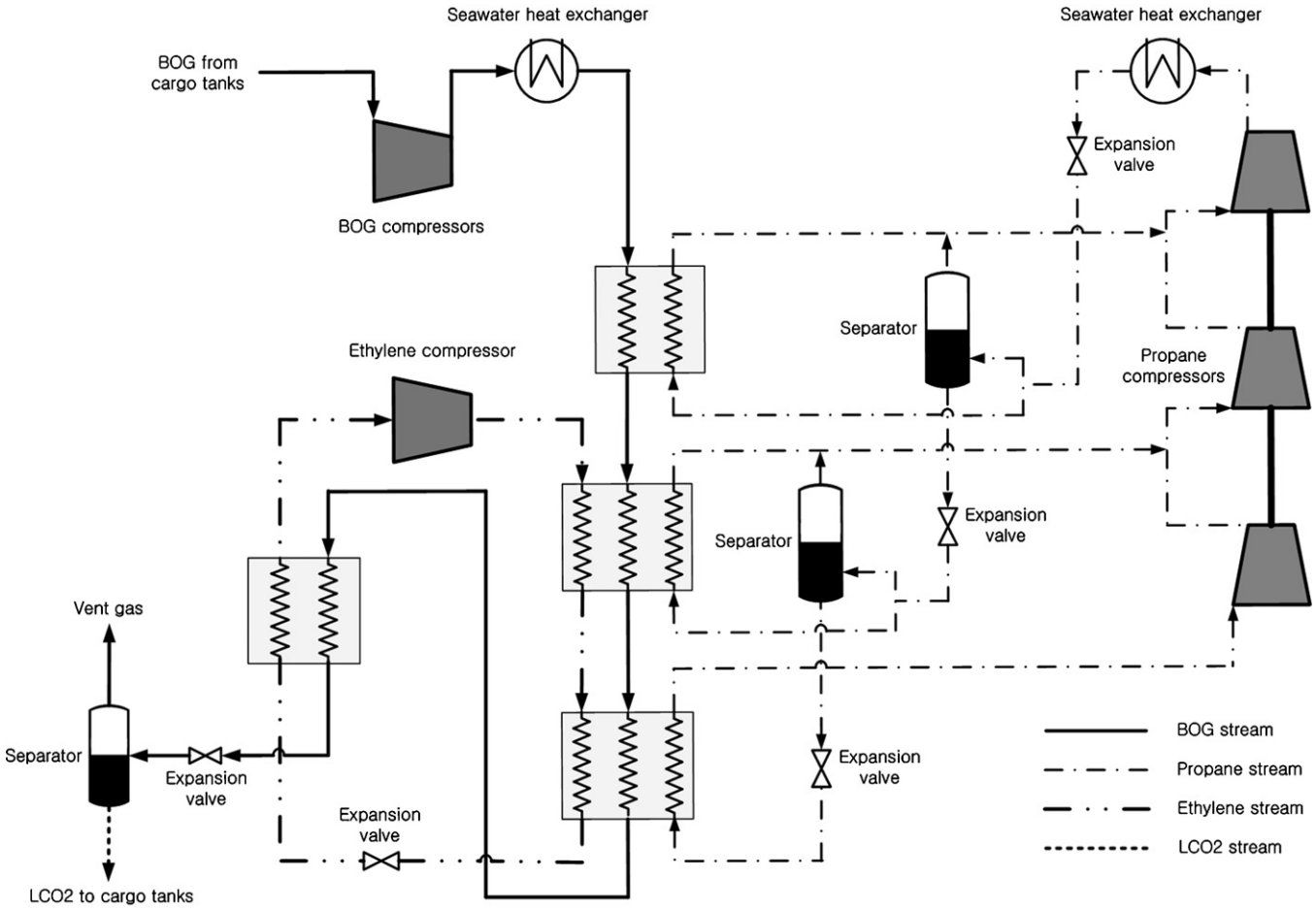


Fig. 3. Process flow diagram for BOG reliquefaction system.

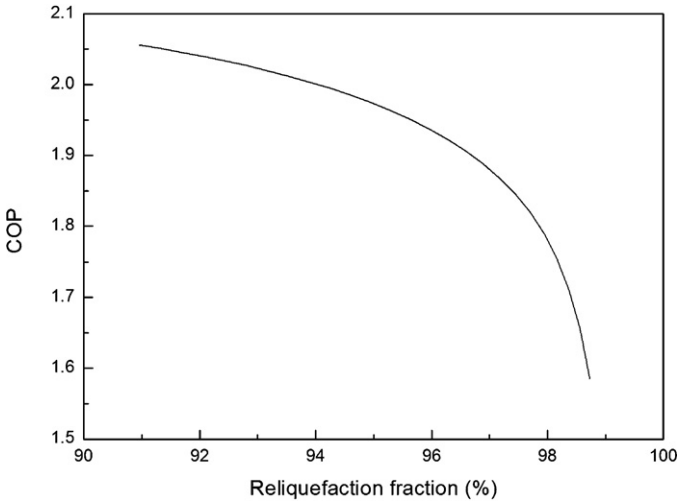


Fig. 4. COP as a function of reliquefaction fraction.

heat to ambient. The propane stream discharged from the expansion valve is split into two depending on heat flow in the heat exchanger. The propane cycle has three cooling and condensation stages to reduce the temperature difference in heat exchangers for improving efficiency. The other refrigerant ethylene is condensed by propane. The ethylene cycle has a condensation stage for liquefying the BOG, which is expanded through a Joule–Thomson valve at the pressure of cargo storage tank, which is 6.5 bar.

The feasibility of the proposed reliquefaction system can be evaluated in terms of the coefficient of performance (COP) (Yunus and Michael, 2006). The COP of the proposed propane–ethylene JT refrigeration system is defined by the following equation:

$$\text{COP} = \frac{\text{heat removed from the BOG}}{\text{total compressor work in the cascade cycle}} \quad (1)$$

The change in the COP of the propane–ethylene JT refrigeration system with evaporating temperature of ethylene is shown in Fig. 4. The COP of the system is closely related to the reliquefaction fraction. As the reliquefaction fraction increases, the COP of the system decreases. In other words, the lower the evaporating temperature of ethylene is, the lower the efficiency becomes. Kilicarslan and Hosoz (2010) demonstrated that the COP of the commercial cascade refrigeration system ranges from 0.6 to 1.2 with the same evaporating temperature. In contrast, the presented refrigeration system has a COP ranging from 1.6 to 2.1 with an evaporating temperature of ethylene between −69 and −46 °C.

3.2. Optimization of the degree of BOG reliquefaction

When the degree of reliquefaction is low, the heavy component, CO_2, is mainly condensed into the liquid phase with the light components of N_2 and O_2 rich in the vapor phase. There is still some amount of CO_2 in the vapor phase. Full liquefaction of the BOG including the light ends is accompanied by a considerable amount of energy. The more these light components are condensed into the liquid phase, the more energy the reliquefaction process consumes. A low extent of reliquefaction, however, means a significant loss of the cargo CO_2. As a consequence, there is an optimum extent of reliquefaction between the recovered CO_2 and the energy requirement.

As mentioned previously, the generated BOG consists of three components, which are CO_2, N_2, and O_2. In evaluating economic feasibility of reliquefaction, CO_2 only affects the total treatment expense as a greenhouse gas. Therefore, BOG refers to only CO_2 in the generated BOG in the following text.

The generated BOG is split into two parts: one that is recovered through reliquefaction and another that is emitted to the atmosphere.

$$\dot{m}_B = \dot{m}_L + \dot{m}_V \quad (2)$$

Here $\dot{m}_B$, $\dot{m}_L$, and $\dot{m}_V$ are the generated, liquefied, and vented BOG, respectively. The reliquefaction fraction is defined by the ratio of the liquefied BOG to the generated BOG.

$$x \equiv \frac{\dot{m}_L}{\dot{m}_B} \quad (3)$$

The vented BOG is expressed in terms of x by the following equation:

$$\dot{m}_V = (1 - x)\dot{m}_B \quad (4)$$

The total treatment expense for the recovery of normal gases can be evaluated by considering the reliquefaction expense and the benefit of the liquefied BOG sale. In the case of handling a greenhouse gas, however, the penalty for the emitted carbon dioxide to the atmosphere should be taken into account. The total treatment expense for a unit mass of BOG, E_T, is a combination of the reliquefaction expense, E_L, the emission penalty, E_P, and the benefit of CO_2 sale, B_S.

$$E_T = E_L + E_P - B_S \quad (5)$$

Generally speaking, the reliquefaction expense, E_L, consists of two parts: the capital expenditure (CAPEX) and the operating expenditure (OPEX). The CAPEX may be deduced from the equipment purchase cost (Peters et al., 2003). Though the capacity of equipment is affected by the reliquefaction fraction, the equipment count is not. Consequently, the CAPEX is a weak function of the reliquefaction fraction. This study assumes that the CAPEX is invariant to the reliquefaction fraction and the reliquefaction expense is determined exclusively by the OPEX, which a direct function of the reliquefaction fraction. The OPEX includes the expense for electricity and operating labor. The electric cost is estimated from the fuel cost for the electric generation. It is strongly dependent on the reliquefaction fraction 'x'. The operating labor, another part of OPEX, is negligible over a range of reliquefaction fractions. Consequently, the reliquefaction expense can be expressed as the total power consumption for liquefying BOG.

$$E_L = \frac{\dot{W}_{tot}(x)}{\dot{m}_B} C_L = w(x) C_L \quad (6)$$

Here $\dot{W}_{tot}(x)$, C_L, and $w(x)$ are the total compressor work, the fuel cost for electric power generation, and the specific compression work, respectively. The total compressor work, $\dot{W}_{tot}(x)$ is a strong nonlinear function of the reliquefaction fraction, increasing steeply with the reliquefaction fraction.

The emission penalty is directly proportional to the emitted CO_2 mass due to the BOG reliquefaction. In addition to the vented BOG, the emitted CO_2 should include the CO_2 mass, $\dot{m}_G$, produced by burning fuel to generate the electric power for the reliquefaction unit. Consequently, the emission penalty is expressed by multiplying the total emitted CO_2 mass and the penalty for unit CO_2 emission, C_P.

$$E_P = \frac{\dot{m}_V + \dot{m}_G}{\dot{m}_B} C_P = \frac{(1 - x)\dot{m}_B + \dot{m}_G}{\dot{m}_B} C_P \quad (7)$$

Note that $\dot{m}_G$ is directly proportional to the total compressor work.

$$\dot{m}_G = k\dot{W}_{tot}(x) \quad (8)$$

The proportional constant k is completely dependent upon the generator efficiency and fuel.As a result, the emission penalty can be

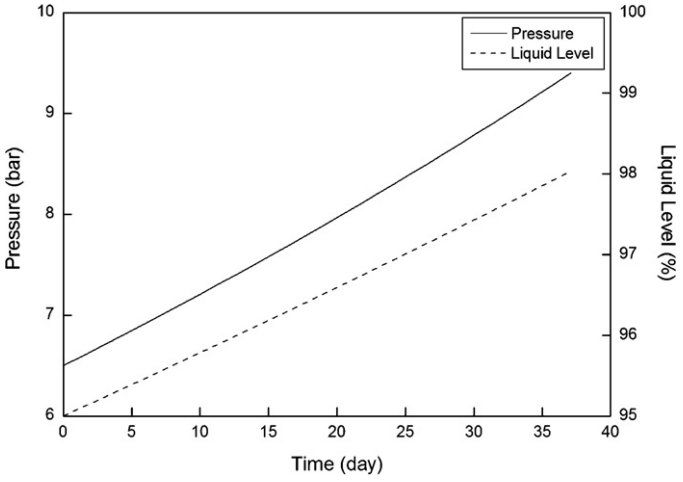


Fig. 5. Evolution of pressure and liquid level of cargo tank inlet without BOG venting.

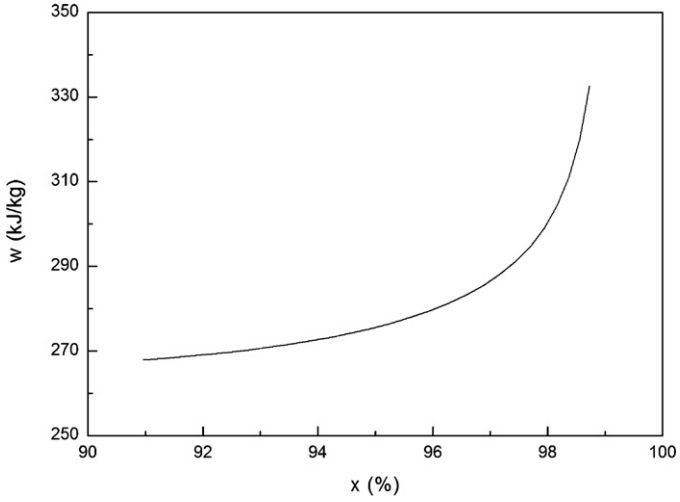


Fig. 6. Specific work with reliquefaction fraction.

obtained by substituting $\dot{m}_G$ from Eq. (8) into Eq. (7), and it is completely a function of the reliquefaction fraction.

$$E_P = \frac{\dot{m}_V + \dot{m}_G}{\dot{m}_B} C_P = \frac{(1-x)\dot{m}_B + k\dot{W}_{tot}(x)}{\dot{m}_B} C_P = [(1-x) + kw(x)]C_P \quad (9)$$

Provided the cargo may be used for commercial purpose such as EOR, the recovered BOG is valuable. For this reason, the benefit of CO_2 sale, B_S, should also be included for evaluating the total treatment expense.

$$B_S = \frac{\dot{m}_L}{\dot{m}_B} C_S = \frac{x\dot{m}_B}{\dot{m}_B} C_S = xC_S \quad (10)$$

Here C_S is the CO_2 sale price in the commercial market.

Based on the aforementioned considerations, the total treatment expense, E_T, is a combination of the reliquefaction expense, the emission penalty, and benefit of CO_2 sale.

$$E_T = w(x)C_L + [(1-x) + kw(x)]C_P - xC_S \quad (11)$$

At the optimum reliquefaction fraction, the derivative of the total treatment expense vanishes.

$$\frac{dE_T}{dx} = \frac{dw(x)}{dx} C_L - C_P + kC_P \frac{dw(x)}{dx} - C_S$$

$$= \frac{dw(x)}{dx}(C_L + kC_P) - (C_P + C_S) = 0 \quad (12)$$

The derivative of the specific work can be expressed by the following equation at the optimum reliquefaction fraction.

$$\frac{dw(x)}{dx} = \frac{C_P + C_S}{C_L + kC_P} \text{ at optimum} \quad (13)$$

If the derivative does not vanish in the domain of [0,1], the optimum reliquefaction fraction lies on the boundary of the range, that is, no reliquefaction ($x = 0$) or full reliquefaction ($x = 1$).

4. Results and discussion

The continuous heat ingress into the cargo tanks results in a steady increase in pressure, temperature, and liquid level. The temperature of the internal atmosphere of the cargo tanks increases with time, as does the pressure. Moreover, the liquid level of the cargo increases due to the thermal expansion of the liquid cargo, ultimately causing irrecoverable damage to the cargo containment system. Fig. 5 shows the increases in the pressure and the liquid level with time for the BOG rate of 0.12%/day without venting the BOG. The internal pressure increases steadily up to 7 bar over 7 days. Considering the design margin of 10% for the pressure vessels, the BOG should be treated (vented or reliquefied) before one week after loading.

A curve of the reliquefaction fraction versus the total compressor work for the BOG reliquefaction unit is shown in Fig. 6. As the reliquefaction rate increases, the specific work increases rapidly because the reliquefaction process expends a considerable amount of energy to condense the light components of N_2 and O_2 at the high reliquefaction fraction.

The operating expenditure is calculated by the cost of fuel oil for electric generation. This study assumes that marine diesel oil (MDO) should be used for diesel generators as in most ships. To predict the MDO price in the future, the correlation between the MDO price and that of crude oil is employed. U.S. Energy Information Administration (2010) provides the future crude oil prices, as shown in Table 4.

The range of the world oil prices is expected to vary from 52 $/barrel to 204 $/barrel in 2030. The future MDO price is derived from historical trend analysis between the crude oil price and the MDO price. Trend line of MDO price with crude oil price is shown in Fig. 7.

Based on the analysis with the crude oil price in 2030 and the trend line, the MDO price is estimated at 1100 $/ton for a 124 $/barrel of crude oil price (currently, 1030 $/ton for a 110 $/barrel of crude oil price). Fig. 8 shows the operating cost with the reliquefaction fraction for this MDO price. Because the operating cost is measured as the specific work multiplied by the fuel price, as shown in Eq. (6), the shape of operating cost line looks similar to the line of the specific work found previously. The operation cost ranges from 14 $/ton-BOG at the reliquefaction fraction of 91% to 17 $/ton-BOG at the reliquefaction fraction of 99%.

As stated previously, the economic feasibility of the reliquefaction system should consider the amount of vented CO_2 and the

Table 4
World oil prices in four cases, 2008–2035 (2008 dollars per barrel) [U.S. Energy Information Administration (2010)].

Year	IEO2010		
	Reference	Low oil price	High oil price
2008	100	100	100
2015	95	52	145
2020	108	52	186
2025	115	52	196
2030	124	52	204
2035	133	51	210

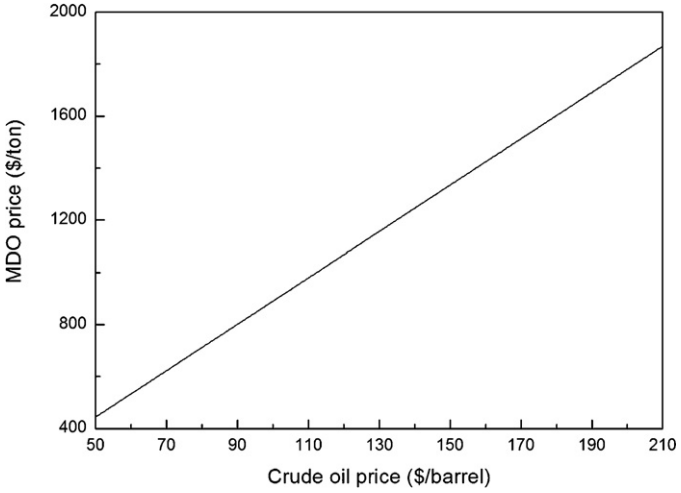


Fig. 7. Trend line of MDO price with crude oil price.

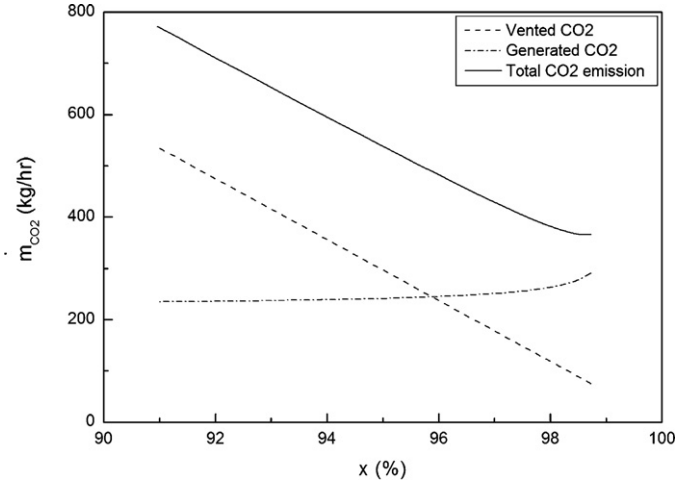


Fig. 9. CO_2 emission with reliquefaction fraction.

operating cost. The amount of CO_2 emission includes not only the vented BOG but also the CO_2 generated from power consumption to reliquefy the BOG. This additional CO_2 emission is estimated from a simple carbon balance for the typical MDO element composition: 86% carbon, 13% hydrogen, and 1% sulfur in mass (Brussen et al., 2006).

A curve of the reliquefaction fraction versus the CO_2 emission to the atmosphere is shown in Fig. 9. As the reliquefaction fraction increases, the amount of the vented BOG decreases. On the contrary, the generated CO_2 from the power generation unit remains nearly constant for the low reliquefaction fraction and increases considerably for the high reliquefaction fraction. As a result, the net CO_2 emission decreases with the reliquefaction fraction.

The penalty for CO_2 emission is estimated through carbon assets in the carbon market. Table 5 shows the trading volumes and values of CO_2 during the period of 2008–2009. The carbon market increased 6% (114 billion $) compared with the previous year, and 8.7 billion tCO_2e were traded in the world. The reason why the specific value of CO_2 declined by 41% (16.5 $/ton) in 2009 is the global economic downturn (Kossoy and Ambrosi, 2010). It is likely that the value will return to the normal level with the global economy recovery.

The future carbon tax is foreseen in three cases: 20, 50, and 100 $/ton in 2030 based on the fourth assessment report of the Intergovernmental Panel on Climate Change (2007). The penalty for CO_2 emission on the basis of the different carbon tax scenarios is shown in Fig. 10. As expected, total penalty increases more rapidly with decreasing reliquefaction fraction in the higher carbon tax scenario.

The recovered BOG is beneficial when the cargo is used for EOR. Currently, the sales price of CO_2 is approximately 20 $/ton in the offshore field (Kapteijn, 2010). The value of CO_2 for EOR is highly correlated with the crude oil price. Holt et al. (2009) estimated that the CO_2 price varies from −23 $/ton-$CO_2$ to 94 $/ton-$CO_2$ for crude oil prices in the range 20 and 160 $/barrel. In this respect, if the oil price increases to 124 $/barrel, the CO_2 price for EOR would increase up to 80 $/ton. As inferred from Eq. (10), the benefit of CO_2 sale is a linear function of the reliquefaction fraction with the slope of the CO_2 price for EOR, as shown in Fig. 11.

The three components of the total treatment expense, which are the reliquefaction expense, the penalty for CO_2 emission, and the benefit of CO_2 sale, are estimated by predicting carbon market in 2030. The total treatment expense for recovering the generated BOG is expressed as a function of the reliquefaction fraction by Eq. (11). On the basis of the different carbon tax scenarios, the total treatment expense with the reliquefaction fraction is shown in Fig. 12.

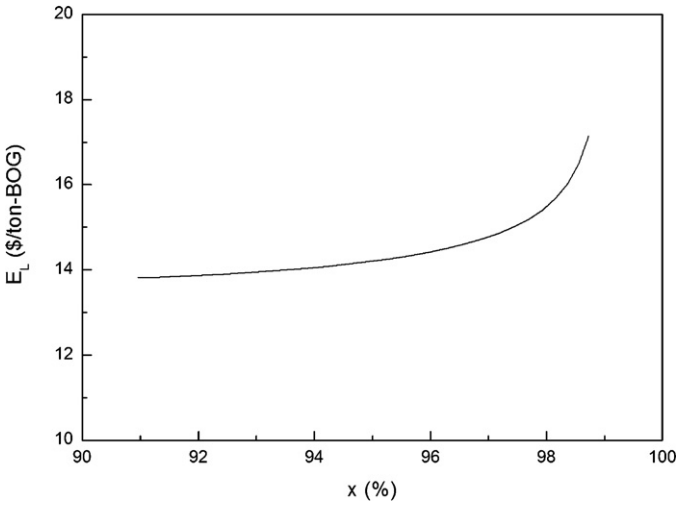


Fig. 8. Operating cost with reliquefaction fraction for 1200 $/ton-MDO.

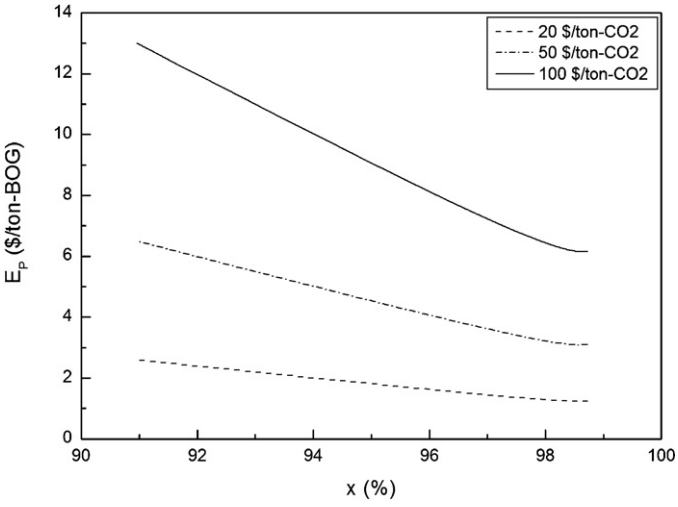


Fig. 10. Penalty for CO_2 emission with reliquefaction fraction.

Table 5
Volumes and values of carbon market in 2008–2009 (Kossoy and Ambrosi, 2010).

	2008		2009	
	Volume ($MtCO_2e$)	Value (million $)	Volume ($MtCO_2e$)	Value (million $)
Allowances markets				
EU ETS	3093	100,526	6326	118,474
NSW	31	183	34	117
CCX	69	309	41	50
RGGI	62	198	805	2179
AAUs	23	276	155	2003
Subtotal	3278	101,492	7362	122,822
Spot and secondary kyoto offsets				
Subtotal	1072	26,277	1055	17,543
Project-based Transactions				
Primary CDM	404	6511	211	2678
JI	25	367	26	354
Voluntary market	57	419	46	338
Subtotal	486	7297	283	3370
Total	4836	135,066	8700	143,735

$MtCO_2e$, million ton carbon dioxide equivalent.

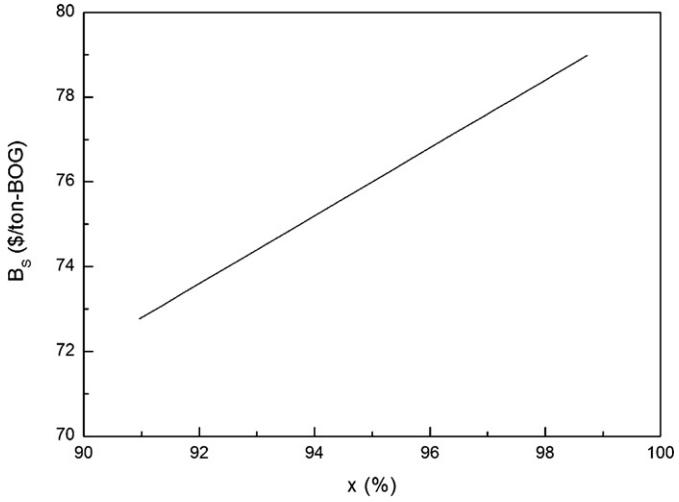


Fig. 11. Benefit of CO_2 sale as a function of reliquefaction fraction.

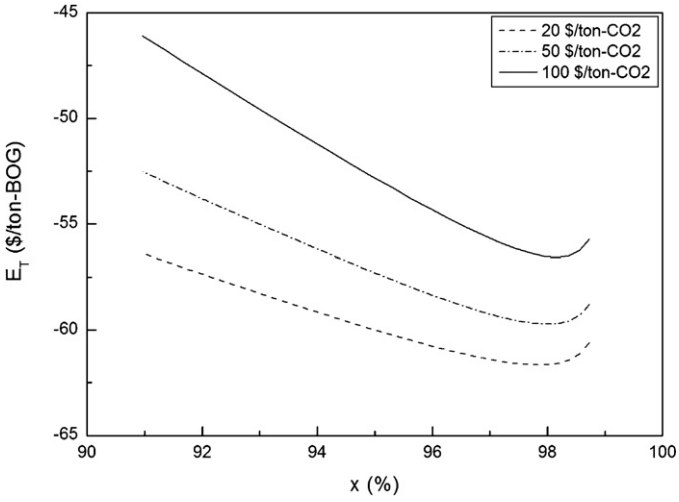


Fig. 12. Total treatment expense with reliquefaction fraction.

The negative sign of the total treatment expense means that the benefit of CO_2 sale is larger than the sum of the reliquefaction expense and the penalty for CO_2 emission. In other words, when the transported CO_2 is used for EOR, BOG reliquefaction results in significant profits. As the carbon tax increases, the total treatment expense decreases with the increasing benefit, and the optimum reliquefaction fraction moves to the higher reliquefaction fraction.

Fig. 13 depicts the derivative of the total treatment expense with the reliquefaction fraction for the three carbon tax scenarios. The optimum liquefaction fraction is given when the derivative is equal to zero. Eq. (13) indicates that the optimum reliquefaction fraction is a function of a combined relationship of C_P, C_S, C_L, and k as well as the derivative of the specific compression work. Even though each varies significantly, the combined relationship does not. The proportional constant k is fixed with the generator efficiency and fuel. The fuel cost (C_L) and the CO_2 sale price (C_S) mainly depend on the crude oil price which is expected to be in the range between the low oil price, 52 $/barrel, and the high oil price, 204 $/barrel, in 2030. In this crude oil price range, the minimum and maximum values of fuel cost are 463 $/ton-MDO and 1816 $/ton-MDO, respectively. The CO_2 sale price will be bounded by 8 $/ton-$CO_2$ and 156 $/ton-$CO_2$ for the future crude oil price range. A range of optimum reliquefaction fractions for the 20 $/ton-$CO_2$ of carbon tax is shown in Fig. 14.

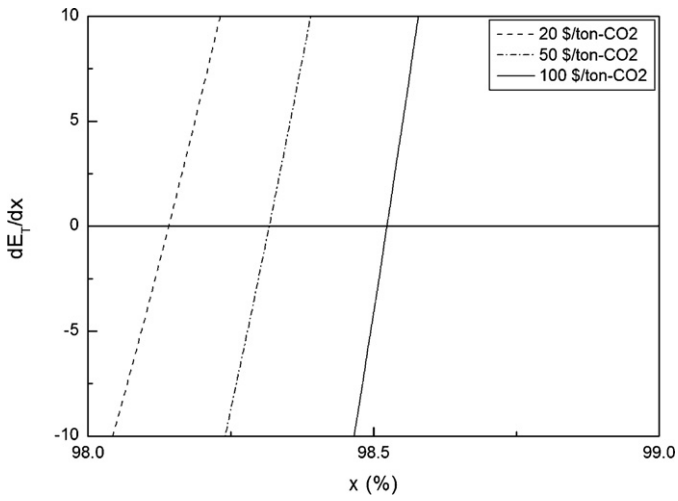


Fig. 13. Optimum reliquefaction fraction with derivative of total treatment expense.

Table 6
Comparison of life-cycle cost of reliquefaction and non-reliquefaction for 20-year life cycle.

	Item	Unit	Carbon tax, $/ton-$CO_2$		
			20	50	100
Reliquefaction case	Profit from BOG treatment	million $/year	2.9	2.8	2.7
	CAPEX (direct cost and indirect cost)	million $		8.9	
	OPEX (excluding electricity)	million $/year		1.3	
	Life-cycle cost	million $	−22.1	−20.3	−17.2
Non-reliquefaction case	Life-cycle cost	million $	18.9	47.3	94.6

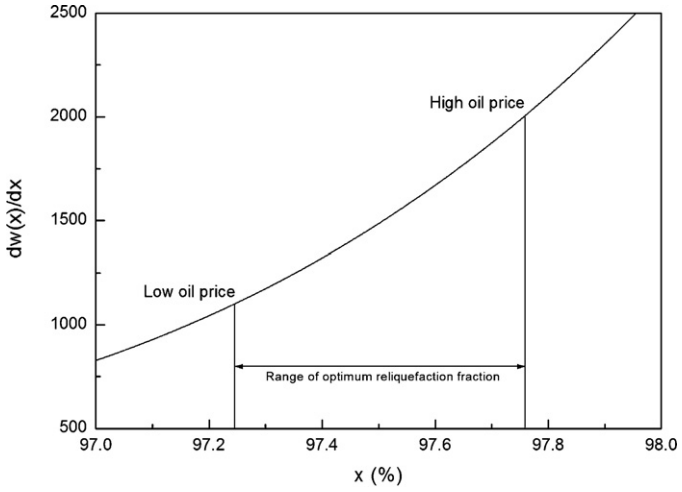


Fig. 14. Range of optimum reliquefaction fraction at 20 $/ton-$CO_2$ of carbon tax.

In the future crude oil price case in the range from 52 $/barrel to 204 $/barrel, the optimum reliquefaction fraction varies only from 97.2 to 97.7%.

Fig. 15 shows a range of optimum reliquefaction fractions in various scenarios of carbon tax and crude oil price. The full extent of the optimum reliquefaction fraction for every scenario including the fuel cost, CO_2 sale price, and penalty for unit CO_2 emission is shown in this graph. In all crude oil prices, the lowest of the optimum reliquefaction fraction appears at 20 $/ton-$CO_2$ of the carbon tax. On the contrary, 100 $/ton-$CO_2$ of the carbon tax has the highest optimum reliquefaction fraction in the entire range of crude oil price. In addition, the lower crude oil price has a larger range of optimum reliquefaction fractions than the higher crude oil price.

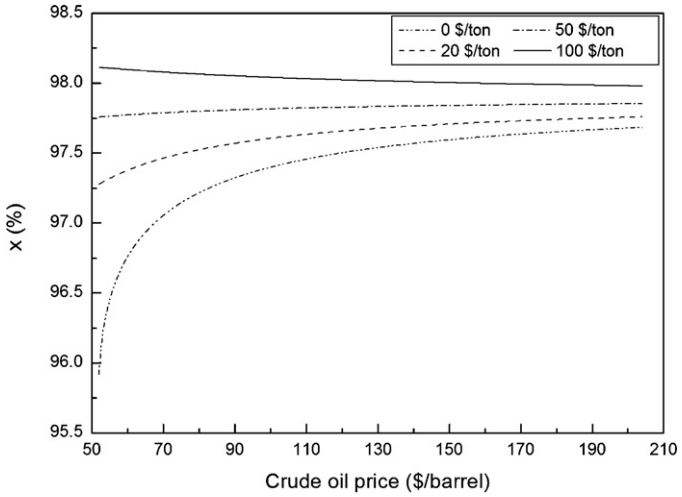


Fig. 15. Range of optimum reliquefaction fraction in three carbon tax scenarios and no penalty case.

Consequently, the maximum value of the optimum reliquefaction fraction is 98.1%, and the minimum value of the optimum reliquefaction fraction is 97.2% over the whole range of future crude oil and carbon tax scenarios. Even when no penalty is imposed on the emission, the optimum reliquefaction fraction remains in the range of 95.7–97.5% for the crude oil scenario.

Even from the view point of life-cycle cost (LCC) which includes CAPEX as well as OPEX, the reliquefaction option is found much more beneficial than the non-reliquefaction one. Table 6 compares the life-cycle cost of the two cases with and without the reliquefaction plant. It is assumed for both cases that the portion of the CO_2-emitting period should be 90% of the total calendar time, resulting in the BOG generation rate of 47,304 ton-BOG/year. The CAPEX and OPEX of the reliquefaction plant are estimated by the method of percentage of delivered equipment cost (Peters et al., 2003), which is available from Aspen HYSYS (2009). Note that the LCC of the reliquefaction case is negative, meaning that this case makes revenue due to the profit from the BOG treatment. To the contrary, that of the non-reliquefaction case is positive, implying expense due to the penalty or carbon tax for the emitted BOG. The LCC of the non-reliquefaction case reaches 95 million $ in the maximum carbon tax scenario.

5. Conclusions

A new BOG reliquefaction unit for semi-pressurized liquid CO_2 carriers was designed in this study, and the optimum reliquefaction fractions for different carbon taxes were reported. The generated BOG was detrimental to the structural integrity of the cargo containment system since it increased the internal pressure of the cargo tanks by 10% within seven days after cargo loading. A propane–ethylene JT refrigeration system was proposed to condense the generated BOG effectively.

The economic feasibility of the reliquefaction unit was estimated by the total treatment expense, which was expressed as the sum of the electric cost to operate the reliquefaction unit, the penalty for the emitted carbon dioxide to the atmosphere, and the commercial profit of the recovered BOG. Since the first and the last were governed by the crude oil price, the total treatment expense was a function of the emission penalty and the crude oil price. The optimum reliquefaction fraction, which minimized the total treatment expense, was estimated from the generalized expression of the total treatment expense. Even though each component cost varied significantly, the optimum reliquefaction fraction stayed in the range of 95.7–98.2% for all the plausible scenarios of the emission penalty and the crude oil price. The analysis based on 20-year life-cycle cost indicated that the reliquefaction case was much more advantageous over the non-reliquefaction case.

Additional factors may affect the validity of this study. The optimum reliquefaction fraction depends considerably on the specific compression work, more specifically, its derivative. Because the work is governed by the process design and equipment performance, the optimum may change for different reliquefaction system designs. The composition of the BOG, especially the level of the light-end components, may change the optimum by altering

the specific compression work. The cargo storage conditions may also cause the optimum to change. For example, if the storage pressure (equivalently, the temperature) is elevated far above the triple point, the specific compression work is considerably reduced. The derivative of the compression work, however, may show an insignificant change with the reliquefaction fraction. As a consequence, it is expected that the change due to uncertainties in these factors would not be significant considering the narrow domain for the optimum reliquefaction fraction.

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